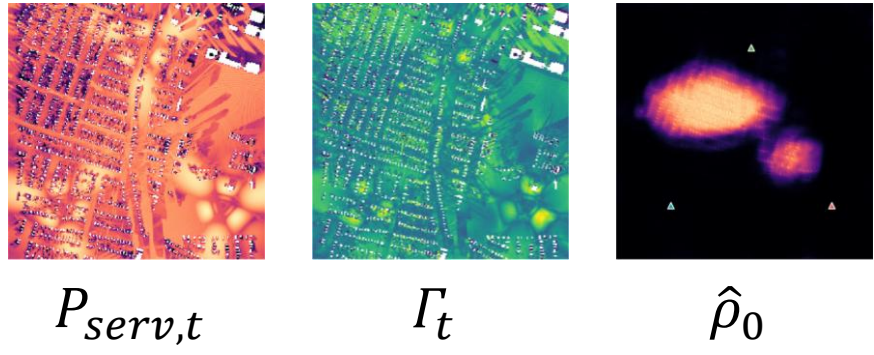


(a) Shared Observation

Network State from digital twin & feedback

Digital Twin Maps & Estimator Result (Current angles θ_t)



UE Reports ($M < N$)

UE	qRSRP	qSINR
...	...	...
M	...	...

TX configuration

θ_1	θ_2	θ_3	...	θ_9	t/H
------------	------------	------------	-----	------------	-----

**Map
Encoder**
 $f_{map}(\cdot)$

**Report
Encoder**
 $f_{set}(\cdot)$

**Config
Encoder**
 $f_{cfg}(\cdot)$

⊕

**Global
Context**
 g_t
256-D

(b) Sector-substate injection

Inject per-sector and demand information

Report substate

Reports grouped for sector k

UE m	Serving qRSRP	Serving qSINR
m_1	$qRSRP_{m_1,k,t}$	$qSINR_{m_1,t}$
m_2	$qRSRP_{m_2,k,t}$	$qSINR_{m_2,t}$
$\vdots$	$\vdots$	$\vdots$
$m_{n_{k,t}}$	$qRSRP_{m_{n_{k,t}},k,t}$	$qSINR_{m_{n_{k,t}},t}$

$u_{k,t}^{rep}$

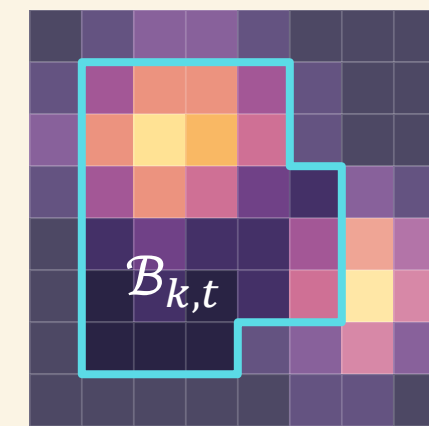
**Sector report
encoder**
 $\psi_{set}(\cdot)$

Learned identity
 e_k

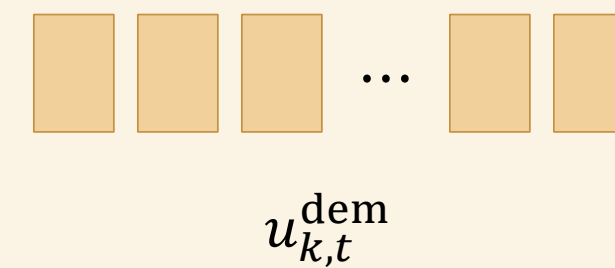
FiLM
 g_t -conditioned

Tokens
 $x_{k,t}$

Demand substate



Demand-weighted radio statistics

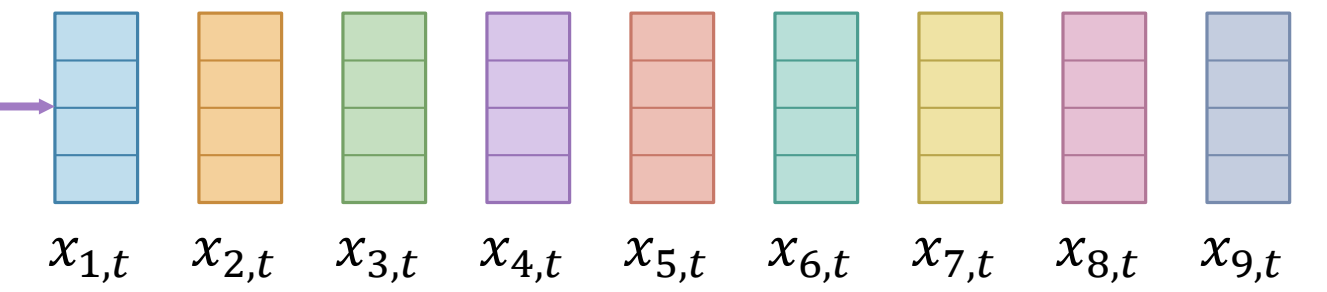


**Demand
Encoder**
 $\psi_{dem}(\cdot)$

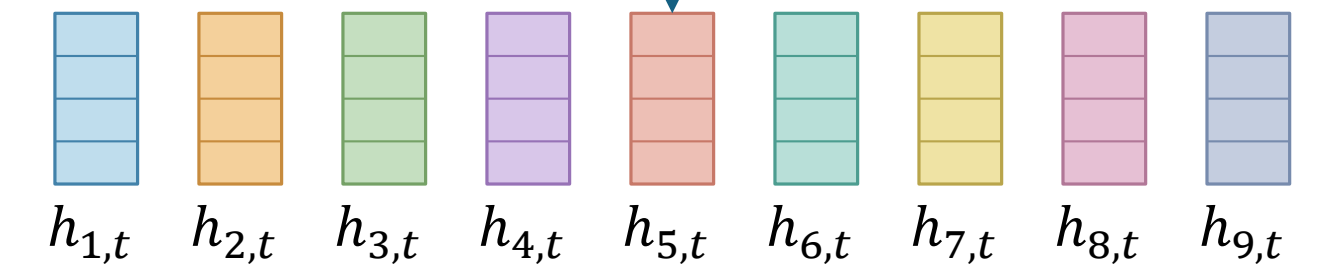
(c) Coordinate and act

One action distribution per sector

9 aligned tokens



Transformer $\mathcal{T}$
Cross-sector attention



Shared policy head
Linear + softmax

Per-sector $\arg\max$

Joint action a_t



$a_{k,t} \in \{-1^\circ, 0^\circ, +1^\circ\}$